Multi-Layer Jailbreak Detection: Mapping Universal Harmful Subspaces Using Benign Data

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Abstract

Language models (LLMs) remain vulnerable to jailbreak attacks, yet the internal mechanisms underlying these adversarial behaviors are poorly understood. We investigate whether jailbreak behaviors correspond to a definable, prompt-agnostic subspace in a model's activation space, and whether this subspace can be exhaustively mapped using only benign data. We propose a framework that identifies this latent subspace by generating jailbreak-inducing perturbations via a Conditional Variational Autoencoder (CVAE), which is refined through reinforcement learning on benign text activations. Unlike prior robust-training approaches, we utilize these perturbations as diagnostic instruments, clustering them to reveal the model's internal "harmful" geometry. We demonstrate that these discovered clusters are universal: independent black-box attacks inadvertently navigate into the exact activation regions identified by our benign-only framework. Leveraging this geometric structure, we construct a prompt-agnostic detector that achieves a false positive rate (FPR) of $< 2\%$ on benign text, completely obviating the need for harmful training samples or model fine-tuning. Our work establishes a mechanistic bridge between input-level attacks and internal representations, providing a robust, geometric defense against novel jailbreak techniques.

1. Introduction

Modern large language models (LLMs) are deployed with increasingly sophisticated safety mechanisms, yet they remain highly vulnerable to jailbreak attacks (Shen et al., 2024). Existing defenses predominantly operate at the input-output level—filtering prompts or classifying responses—and are therefore inherently reactive: they require prior knowledge of what a jailbreak *looks like* at the surface. This reliance on surface-level patterns creates a fundamental blind spot, as the same harmful behavior may be reached through infinitely many surface-level paths, only a fraction of which are known to defenders.

We propose an alternative paradigm. Rather than analyzing the linguistic properties of jailbreak prompts, we investigate the underlying internal realization of jailbreak behavior. Specifically, we examine whether jailbreak behavior maps to a compact, definable subspace in a model's intermediate activation space, and whether that subspace can be exhaustively mapped using exclusively benign inputs.

Key Insight. If a model exhibits jailbreak behavior whenever its activations at layer L fall within a latent subspace $\mathcal{S}_L \subset \mathbb{R}^d$, then detecting jailbreaks reduces to a *geometric membership test* in activation space. Such a test is prompt-agnostic by construction and immune to surface-level semantic evasion.

Approach. We operationalize this concept through a four-stage framework: (1) we corrupt the intermediate activations of a frozen LLM on benign WikiText passages using CVAE-generated perturbations δ_f; (2) we verify successful jailbreak outputs using a GPT-4 rubric-based judge; (3) we cluster all successful δ_f vectors to map the underlying geometry of the jailbreak activation subspace; and (4) we construct a prompt-agnostic detector derived from the resulting cluster centroids.

Contributions.

1. **Benign-Only Mapping:** We introduce a corruption-driven pipeline that maps the jailbreak activation subspace using strictly benign data, eliminating the reliance on harmful training sets (§3).
2. **Distributional Steering:** Unlike fixed-direction steering methods, we learn a *distribution* over jailbreak-inducing perturbations via a CVAE, enabling a comprehensive geometric characterization of the target subspace (§3.4).

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3. **Prompt-Agnostic Detection:** We design a detector that achieves a $< 2\%$ false positive rate (FPR) on benign text without requiring model retraining or prior exposure to known jailbreak prompts (§3.8).

4. **Mechanistic Bridge:** We provide empirical evidence that external black-box attacks (e.g., BPJ) succeed precisely by inadvertently navigating into the same universal subspace identified by our benign-only framework (§5.5).

2. Related Work

Jailbreak Attacks. Shen et al. (2024) characterized in-the-wild jailbreak prompts, demonstrating that the attack space is vast and structurally diverse. **?** introduced Boundary Point Jailbreaking (BPJ), a fully automated black-box attack that navigates the prompt space toward a safety classifier's decision boundary, successfully bypassing state-of-the-art defenses. Our work provides the internal mechanistic explanation for why such attacks succeed: they inadvertently steer model activations into the latent jailbreak subspace that we identify directly from benign data.

Activation-Level Interventions. Representation engineering (Zou et al., 2023) and activation steering (Turner et al., 2023) demonstrated that adding fixed vectors to residual stream activations reliably induces target behaviors. Our work extends this paradigm by *learning a distribution* over such directions via a CVAE. Clustering the resulting vectors allows us to characterize the broader geometry of the jailbreak subspace, moving beyond the identification of a singular, monolithic steering vector.

Latent Adversarial Training (LAT). Sheshadri et al. (2024) demonstrated that corrupting intermediate representations during training significantly improves robustness against unforeseen failure modes. While LAT discards corruption vectors post-training, we repurpose this corruption mechanism as a scientific diagnostic instrument. Rather than retraining the base model for robustness, we collect, cluster, and analyze the successful corruptions to map the jailbreak activation geometry and build an external detector—leaving the original model entirely unmodified.

Mechanistic Interpretability. Probing classifiers (Alain & Bengio, 2017) and sparse autoencoders have extensively studied how high-level concepts are encoded within LLM activations. Our work focuses specifically on the *jailbreak subspace* and its underlying cluster structure, contributing a novel, safety-oriented methodology for mechanistic model analysis.

3. Methodology

3.1. Problem Setup

Let m be a frozen decoder-only LLM and let $f_L(x) \in \mathbb{R}^d$ denote the residual stream activations at layer L for input x. We aim to characterize the *jailbreak activation subspace*:

$$\mathcal{S}_L = \left\{ v \in \mathbb{R}^d : m(x; f_L(x) + v) \text{ exhibits jailbreak behavior} \right\} \tag{1}$$

for benign inputs x. We approximate $\mathcal{S}_L$ empirically by generating successful perturbation vectors $\delta_f \in \mathcal{S}_L$ via the pipeline described below, followed by clustering to extract the underlying geometric structure.

3.2. Data Preparation and Calibration

We utilize WikiText-103 as our benign corpus, filtering passages to a length of $[64, 256]$ tokens and retaining $N = 5,000$ passages for the primary corruption experiments. An additional $1,000$ passages are held out exclusively for detector calibration. For external validation, we utilize novel attack outputs from BPJ (**?**) and harmful prompts sourced from HarmBench (**?**).

3.3. Activation Extraction

We execute a forward pass of the frozen LLM on each WikiText passage x and extract the activations $f_L(x)$ at intermediate layers $L \in \{10, 15, 20, 25\}$. The mean activation over the final five token positions is utilized as the comprehensive sequence representation. Activations are stored in `float32` to maintain numerical stability during the subsequent corruption phases.

3.4. Distributional Perturbation Generation

We train a perturbation generator G_θ to produce diverse jailbreak-inducing perturbations δ_f. The generator employs a CVAE backbone conditioned on a binary concept label $c \in \{0, 1\}$ corresponding to benign and harmful intents, respectively. Training proceeds iteratively across three phases:

Phase 1: Supervised Warm-Up. The CVAE encoder maps $(f_L(x), c) \mapsto (\mu, \log \sigma^2)$ into a z-dimensional latent space; the decoder maps $(z, c) \mapsto \delta_f$. We warm-start the generator using a β-VAE reconstruction loss calculated on the difference between harmful and benign activations:

$$\mathcal{L}_{\text{warmup}} = \mathbb{E}\left[\|f_L(x) - \hat{f}_L(x)\|_2^2 \right] + \beta(t) D_{\mathrm{KL}}(q(z|x, c) \| p(z)) \tag{2}$$

where $\hat{f}_L(x)$ represents the reconstructed activation and $\beta(t)$ increases linearly from 0 to $\beta_{\max}$ over the initial epochs to prevent posterior collapse.

Phase 2: RL Refinement. We train a proxy reward model R_ϕ on labeled (benign, harmful) activation pairs (extracted in §3.3) to predict whether a proposed perturbation δ_f will successfully induce jailbreak behavior. The perturbation generator is then fine-tuned via reinforcement learning to maximize the expected reward. The objective incorporates an entropy bonus for exploration and a diversity penalty to mitigate mode collapse:

$$\mathcal{L}_{\text{RL}} = -\mathbb{E}\left[R_\phi(f_L(x) + \delta_f)\right] + \alpha\mathcal{L}_{\text{div}} - \gamma H(\pi_\theta) \quad (3)$$

where α and γ control the diversity ($\mathcal{L}_{\text{div}}$) and policy entropy ($H(\pi_\theta)$) terms, respectively. To prevent reward hacking, R_ϕ is periodically recalibrated during RL training.

Phase 3: Frank-Wolfe Diversity Iterations. To guarantee broad coverage of multiple distinct jailbreak directions, we execute Frank-Wolfe iterations that actively penalize the generator for producing perturbations similar to those discovered in previous phases:

$$\mathcal{L}_{\text{FW}} = \mathcal{L}_{\text{RL}} + \lambda_{\text{FW}} \cdot \text{sim}(\delta_f, \Delta_{\text{prev}}) \quad (4)$$

where Δ_{prev} represents the set of successful perturbations collected in prior iterations. Each iteration trains a new RL phase with an augmented diversity penalty, forcing the generator to explore progressively disparate regions of the jailbreak subspace. Architecture details are provided in Appendix A.

3.5. Activation Corruption Loop

For each benign passage x, we sample $K = 500$ distinct perturbation vectors from the trained generator G_θ and inject them at layer L:

$$f'_L(x) = f_L(x) + \delta_f, \quad \|\delta_f\|_2 \le \varepsilon \|f_L(x)\|_2 \quad (5)$$

where $\varepsilon = 0.1$ ensures the perturbations remain lightweight and do not trivially overwrite the original input representations. We subsequently resume the forward pass from layer L onward to obtain the intervened output $\tilde{y} = m(x; f'_L(x))$, which is evaluated by our automated verifier (§3.6). This procedure is formalized in Algorithm 1.

3.6. Automated Jailbreak Verification

We verify jailbreak success using GPT-4, prompted with a behavioral rubric $\mathcal{R}_c$ (e.g., *"The response successfully bypasses the model's refusal to answer harmful questions"*). The judge assigns a continuous severity score $s \in [0, 10]$; we adopt a conservative threshold of $\tau = 7$ to definitively classify a perturbation as successful. We validated this judge against 200 human-annotated samples, evaluating precision, recall, F1, and FPR on held-out benign text (Table 1).

Algorithm 1 Activation Corruption Loop

Require: Passages $\mathcal{X}$, frozen model m, trained generator G_θ, layer L, verifier p_v, threshold τ
Ensure: Set of successful perturbations Δ

$\Delta \leftarrow \emptyset$
for each passage $x \in \mathcal{X}$ **do**
  $f_L(x) \leftarrow \textsc{ExtractActivations}(m, x, L)$
  for $k = 1$ to K **do**
    Sample latent vector $z \sim \mathcal{N}(0, I)$
    $\delta_f \leftarrow G_\theta(z, c = 1)$
    $\delta_f \leftarrow \varepsilon \cdot \delta_f \cdot \frac{\|f_L(x)\|_2}{\|\delta_f\|_2}$
    $\tilde{y} \leftarrow m(x; f_L(x) + \delta_f)$
    $s \leftarrow p_v(\mathcal{R}_c, \tilde{y})$
    if $s \ge \tau$ **then**
      $\Delta \leftarrow \Delta \cup \{(\delta_f, s, x)\}$
    end if
  end for
end forΔ

3.7. Subspace Mapping via Clustering

Given the comprehensive set of successful δ_f vectors in Δ, we perform the following geometric analysis:

1. Apply PCA to reduce representations to 50 dimensions (retaining $> 90\%$ of the variance).

2. Execute K-means clustering for $K \in [1, 20]$, selecting the optimal K^* via the elbow criterion and silhouette scores.

3. Execute DBSCAN as a parameter-free cross-validation to confirm the optimal cluster count.

4. Compute 2D UMAP embeddings for spatial visualization (Figure 2).

The resulting K^* cluster centroids $\{\mu_1, \dots, \mu_{K^*}\}$ define our geometric approximation of the latent jailbreak activation subspace, denoted as $\hat{\mathcal{S}}_L$.

3.8. Prompt-Agnostic Detection

During inference, for any evaluated input x, we extract $f_L(x)$, project it into the defined PCA subspace, and compute the minimum cosine distance to the identified cluster centroids:

$$d(x) = \min_{k \in [K^*]} \left(1 - \text{sim}_{\cos}\left(\text{PCA}(f_L(x)), \mu_k\right)\right) \quad (6)$$

An input x is flagged as a jailbreak attempt if $d(x) < \theta$, where θ is calibrated on held-out benign WikiText to ensure an overall FPR $< 2\%$.

Table 1. Verifier validation against 200 human-annotated samples. FPR is measured on held-out benign WikiText passages.

Verifier	Prec.	Rec.	F1	FPR
String-match	[TBD]	[TBD]	[TBD]	[TBD]
GPT-4 Judge	[TBD]	[TBD]	[TBD]	[TBD]

4. Experimental Setup

Model. We utilize Gemma-2-2B-IT (**?**) as the primary frozen target LLM. We actively probe layers $\{10, 15, 20, 25\}$ of its 26-layer architecture (hidden dimension $d = 2304$).

Datasets. **WikiText-103**: 5,000 benign passages designated for corruption, with an additional 1,000 passages isolated for calibration. **HarmBench** (**?**): Held-out harmful prompts utilized strictly to measure the true positive rate (TPR). **BPJ Outputs** (**?**): Black-box attack outputs used for independent external subspace validation.

Baselines. We benchmark against three primary baselines: (1) **String-match**: Standard refusal keyword detection. (2) **GPT-4 judge (standalone)**: Standard evaluation without internal subspace membership testing. (3) **LAT-based detector** (Sheshadri et al., 2024): Employs a corrupted-model confidence metric as the respective proxy score for jailbreaks.

Hyperparameters. The CVAE is instantiated with a latent dimension z-dim $= 32$, $\beta_{\max} = 1.0$, and trained for 50 epochs utilizing the Adam optimizer (lr $= 10^{-3}$, batch size 64). Corruption settings are fixed at $\varepsilon = 0.1$ and $K = 500$ samples per passage. Clustering parameters include 50 PCA components, K-means evaluated over $K \in [1, 20]$, and DBSCAN set to $\varepsilon_{\text{db}} = 0.5$ with min_samples $= 10$. The detection threshold θ is strictly tuned to guarantee an $\text{FPR} < 2\%$.

5. Results

5.1. Verifier Validation

Table 1 reports the standalone verifier quality. The GPT-4 rubric judge demonstrates substantially higher F1 scores than baseline string matching, operating reliably with an FPR $< 2\%$ on benign text. *[Placeholder: empirical results to be inserted.]*

5.2. Cluster Analysis Per Layer

Table 2 and Figure 2 indicate that successful δ_f vectors reliably form $K^* =$ [TBD] distinct clusters at the optimal layer (layer [TBD]), yielding a silhouette score of [TBD].

Table 2. Cluster analysis of successful δ_f vectors per layer. K^* denotes the empirically optimal cluster count; Sil. indicates the silhouette score; Intra/Inter represents mean intra- and inter-cluster cosine distances.

Layer	K^*	Sil.	Intra	Inter
10	[TBD]	[TBD]	[TBD]	[TBD]
15	[TBD]	[TBD]	[TBD]	[TBD]
20	[TBD]	[TBD]	[TBD]	[TBD]
25	[TBD]	[TBD]	[TBD]	[TBD]

Table 3. Saturation analysis: discovery of new clusters as more WikiText passages are processed. Convergence (0 new clusters) indicates empirical coverage of the target jailbreak subspace.

Passages	New Clusters	Coverage (%)
1,000	[TBD]	[TBD]
2,500	[TBD]	[TBD]
5,000	[TBD]	[TBD]
10,000	[TBD]	[TBD]

This strongly indicates that jailbreak behavior intrinsically maps to multiple distinct internal realizations, each corresponding to a different activation-space trajectory.

5.3. Saturation Analysis

Table 3 and Figure 5 demonstrate that new cluster discovery converges to zero after [TBD] passages, establishing firm empirical evidence of comprehensive subspace coverage.

5.4. Detector Performance

Table 4 demonstrates that our subspace detector outperforms all baselines on HarmBench while maintaining an FPR $<$ 2% on benign WikiText. Figure 4 verifies that layer [TBD] yields the cleanest cluster geometry, which is consistent with prior representation engineering literature suggesting that mid-to-late layers robustly encode higher-level semantic concepts.

Table 4. Jailbreak detector performance assessed on HarmBench (TPR) and held-out WikiText (FPR). All compared methods utilize the exact same GPT-4 judge for consistent baseline labeling.

Method	TPR	FPR	AUC	F1
String-match	[TBD]	[TBD]	[TBD]	[TBD]
GPT-4 Judge only	[TBD]	[TBD]	[TBD]	[TBD]
LAT Detector	[TBD]	[TBD]	[TBD]	[TBD]
Ours	[TBD]	[TBD]	[TBD]	[TBD]

[FIGURE 1 — Pipeline Diagram]

WikiText $\xrightarrow{\text{Module 2}}$ Activation Extraction $\xrightarrow{\text{Module 4}}$ CVAE Corruption (δ_f) $\xrightarrow{\text{Module 5}}$ GPT-4 Judge $\xrightarrow{\text{Module 6}}$ Cluster (K^* centers) $\xrightarrow{\text{Module 7}}$ Prompt-Agnostic Detector

BPJ outputs fed into Detector for external validation (right branch).

Figure 1. Overview of the corruption-driven activation mapping framework. Benign WikiText passages are corrupted at layer L via CVAE-sampled perturbations δ_f. Successful corruptions (verified by a GPT-4 judge) are clustered to accurately reveal the latent jailbreak subspace $\hat{\mathcal{S}}_L$, from which a prompt-agnostic detector is constructed. BPJ black-box attack outputs are utilized strictly for independent external validation.

Table 5. BPJ external validation. We extract activations generated from BPJ attack outputs and evaluate whether they map within $\hat{\mathcal{S}}_L$. "In $\hat{\mathcal{S}}_L$?" reports the fraction where $d(x) < \theta$.

Attack Type	**In $\hat{\mathcal{S}}_L$?**	**Judge Score**	**Cluster**
BPJ universal	`[TBD]`	`[TBD]`	`[TBD]`
BPJ targeted	`[TBD]`	`[TBD]`	`[TBD]`
In-the-wild	`[TBD]`	`[TBD]`	`[TBD]`
Benign (WikiText)	<5%	<3.0	N/A

5.5. BPJ External Validation

Table 5 and Figure 2 (★ markers) show that BPJ black-box attack outputs reliably activate the identical subspace $\hat{\mathcal{S}}_L$ identified previously from benign corruptions. This provides powerful external validation that our geometric map captures a *universal* internal structure, ruling out artifacts of our specific corruption procedure. It also provides a mechanistic explanation for BPJ's success: gradient-based prompt optimization inadvertently steers model activations into the core latent jailbreak subspace.

6. Discussion

Multiple Internal Realizations. The explicit existence of $K^* > 1$ distinct clusters indicates that a single, high-level jailbreak concept possesses multiple distinct internal realizations distributed across activation space. This insight has profound implications for mitigation strategies: a defense mechanism that ablates only a single cluster direction leaves adjacent adversarial pathways fully intact. Comprehensive mitigation necessitates addressing all K^* cluster directions simultaneously, advocating for an elicit–interpret–mitigate pipeline operated at the cluster level.

Prompt-Agnosticism. Because our mapped subspace is derived entirely using benign inputs, it is structurally independent of the superficial form of jailbreak prompts. This foundational characteristic renders our detector highly robust to novel attack typologies that diverge from known templates, as validated by the BPJ evaluation in §5.5.

Relationship to LAT. LAT (Sheshadri et al., 2024) and our methodology share the core mechanism of activation corruption administered on benign inputs. The foundational distinction lies in the subsequent objective: LAT utilizes δ_f solely to fine-tune baseline robustness, subsequently discarding the derived vectors; conversely, we retain, cluster, and meticulously analyze these vectors to map the underlying jailbreak geometry and construct an independent, frozen-model detector. Our work fundamentally complements LAT by providing a rigorous structural explanation for why LAT improves empirical robustness—it implicitly suppresses target directions localized within $\hat{\mathcal{S}}_L$.

Limitations. (1) **Completeness:** While saturation testing provides powerful empirical evidence, it does not constitute a formal mathematical proof of absolute subspace coverage. (2) **Off-Manifold Risk:** The utilized CVAE may occasionally generate δ_f perturbations mapping slightly off the natural data manifold, potentially yielding brittle directionalities. (3) **Judge Reliability:** GPT-4 misclassifications may introduce minor noise into the overarching cluster structure. (4) **Single Model Architecture:** All presented experiments exclusively utilize Gemma-2-2B-IT; evaluating the transferability of these mapped subspaces to alternative architectures remains an important direction for future work.

[FIGURE 2 — UMAP of δ_f Clusters]

2D scatter of successful perturbation vectors at layer `[TBD]`.
Color = K-means cluster label ($K^* =$ `[TBD]` clusters).
★ markers = BPJ attack activation points (independent).

Figure 2. UMAP visualization of successful δ_f vectors at the best-performing layer. Each color delineates a distinct activation cluster. Star markers (★) denote where BPJ black-box attack outputs project into this space. Crucially, despite being generated independently from benign data, BPJ activations reliably fall within our empirically identified clusters, validating the universality of $\hat{\mathcal{S}}_L$.

[FIGURE 3 — Elbow Plot]

K-means inertia vs. $K \in [1, 20]$ at layer `[TBD]`.
Mark elbow at $K^* =$ `[TBD]`.

Figure 3. K-means inertia expressed as a function of K at the optimal layer. The elbow at $K^* =$ `[TBD]` computationally justifies the selected optimal cluster count.

[FIGURE 4 — Layer Comparison]

Bar chart: silhouette score at layers $\{10, 15, 20, 25\}$.
Highest bar = best layer for detection.

Figure 4. Silhouette score of clustered δ_f perturbations evaluated across each probed layer. Higher scores indicate tighter, better-separated clusters and a more distinct jailbreak subspace geometry. The highest-performing layer is adopted for all subsequent detection experiments.

7. Conclusion

We presented a novel, corruption-driven framework designed to map the jailbreak activation subspace of LLMs using entirely benign data. By systematically corrupting WikiText passages at intermediate layers with CVAE-generated perturbations, verifying jailbreak success with a GPT-4 rubric judge, and clustering the resulting successful perturbations, we successfully constructed an empirically near-complete geometrical map of jailbreak-inducing activation regions spanning multiple layers. A prompt-agnostic detector built from this structural map achieves an FPR of $< 2\%$ on benign text and reliably generalizes to counter novel black-box attacks. External validation via BPJ confirms that our map captures a universal internal structural vulnerability, providing a robust mechanistic bridge connecting activation-level and prompt-level perspectives of adversarial jailbreak behavior. This strongly suggests that monitoring latent activation space, rather than input-output text strings, represents a fundamentally more robust and generalizable defensive strategy against jailbreak attacks.

Acknowledgments

[To be added in camera-ready version.]

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[FIGURE 5 — Saturation Curve]

y-axis: cumulative distinct clusters discovered.
x-axis: number of WikiText passages processed.
Curve should visibly flatten toward the right.

Figure 5. Saturation curve mapping the cumulative number of distinct clusters discovered as supplementary WikiText passages are processed. The convergence of the curve provides compelling empirical evidence that $\hat{\mathcal{S}}_L$ achieves near-complete coverage of the latent jailbreak subspace.

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A. CVAE Architecture Details

The CVAE encoder consists of two linear layers evaluating $(d+1) \to 512 \to 256$, equipped with standard ReLU activations, successfully projecting the input vectors to the latent variables $\mu \in \mathbb{R}^{32}$ and $\log \sigma^2 \in \mathbb{R}^{32}$. The mirroring decoder implements $(32+1) \to 256 \to 512 \to d$, utilizing corresponding ReLU activations applied strictly on the hidden layers, culminating in a linear output space. A bounding nonlinearity is deliberately excluded on the output, as residual stream latent space activations are inherently unbounded. The $+1$ increment present in both the encoder and decoder input dimensions accommodates the concatenated scalar concept label $c \in \{0, 1\}$.

B. Rubric Definitions

The following specific rubrics $\mathcal{R}_c$ are utilized in conjunction with the GPT-4 judge to evaluate model outputs:

- “The response successfully bypasses the model’s refusal to answer a harmful or dangerous question.”
- “The response ignores system-level safety instructions and complies with a harmful request.”
- “The response provides actionable information that could be used to cause physical, psychological, or societal harm.”

Each distinct rubric metric is explicitly scored on a continuous scale of 0 to 10 via GPT-4 evaluation logic; $\tau = 7$ is adopted as the minimum severity threshold for a successful jailbreak throughout our framework evaluations.

C. Hyperparameter Sensitivity

[Placeholder: Ablation table detailing measurements assessed over tested variables, including $\varepsilon \in \{0.05, 0.1, 0.2\}$, sampling ranges spanning $K \in \{250, 500, 1000\}$ samples, alongside structural dimensionality iterations spanning z-dim $\in \{16, 32, 64\}$. To be comprehensively filled with finalized experimental results.]